

Media Speedia

There are quite a few DVD burners available now that support write speeds of up to 16X. However, availability of 16X DVD media is still a big problem, even outside India. In fact, branded 8X media have just started selling here. We had quite a few problems sourcing branded media for this test, and were actually saved by one of our colleagues who got us branded media from the United States. The media we used are the 8X Verbatim DVD+R and the 8X Memorex DVD-R, both of which are capable of writing at up to 12X speeds. After spending almost Rs 10,000 to purchase the blank media required for this test, the Digit team can proudly claim to have some of the most expensive coasters!

came with stands for vertical mounting. Western Digital also supplied rubber pegs, to safeguard the body from scratches.

USB 2.0 was the interface of choice. Since many graphics professionals prefer Macintosh systems to run their graphical tools, FireWire was supported. The IOmega, however, didn't support FireWire.

Disk-side USB connectors used in hard drives were standard, unlike in digital cameras where vendors modify their shape—which forces users to buy proprietary USB cables at premium rates. All the drives came with the standard USB connector, cables for which are readily available in the market.

Another great development of late, in external hard drives, is the application launch button on the device itself. The button launches the backup software provided with the drive. Although not a revolutionary feature, it saves a few mouse clicks.

Maxtor called this button the One Touch, Seagate called it Backup, whereas Freecom called it SYNC. Western Digital went one step ahead and incorporated 2 buttons—one each for automatic and manual backups.

All five drives came with good software. The Freecom FHD-1 120GB came with Acronis True Image, which can create an image,



The Western Digital Media Center

restore data, write an image to a CD or DVD, and also restore from a network.

Iomega came with Norton Ghost 2003; the Maxtor and Western Digital were bundled with Dantz Retrospect Express, which is capable of locating the files to be backed up on the media without user intervention. It can even create restore points.

Seagate provided BounceBack Express, which can restore lost data, and back up data from any mapped drive on the network.

It would be an understatement to call the Western Digital Media Center an external hard drive. It is rightly named "Media Center," because it not only functions as external hard drive, but also has an 8-in-1 card reader and two extra USB and one extra FireWire port—which makes it a USB/FireWire hub.

Freecom and Seagate were the only other drives that had extra FireWire ports. The IOmega came with no extra ports.

Build Quality

Since external drives house very sensitive insides, the package has to be sturdy enough to take the shocks and jerks associated with a portable medium. It also has to be splash- and dust-resistant for smooth operation over a long period of time.

All the drives featured very good build quality, with Freecom topping the chart. The Freecom FHD-1 120 GB had a tiny fan at the back to regulate the heat generated by its 7,200-rpm drive.

Seagate and Western Digital, on the other hand, had a cage design for air regulation, which also exposed the drive and its electronics to dust and water—not a very good idea. None of the drives came with a carry case.

Performance

We performed three sets of tests to gauge the performance of the drives. We ran two synthetic benchmarks—HDTach and SiSoft Sandra 2004 Pro, to get the data transfer speed, access time and CPU utilisation.

In SiSoft Sandra, we got slayer DVD writers, dual-layer DVD writers and external DVD writers—the maximum number of drives were single-layer, internal DVD writers.

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DVD writers. We were glad to see eight drives from seven difecom and Maxtor One Touch managed a respectable 31 MBps in the sequential read test. This means that these drives will take the least time to restore a drive image, which is a single, large file. In the random write test, the Maxtor One Touch reached a high of 17 MBps, which is a good score for an external drive.

In HDTach 2.7, the Freecom FHD-1 120GB drive returned decent scores. It logged an access time of 13.3 compared to the 21.2 logged by Maxtor, because of its multiple platters and heads.

The Seagate ST3160024A 160GB edged passed the Freecom by a few decimal points in average write speed, logging 18.3 MBps. The sore point in this test was the CPU utilisation, which was in the range of 28 to 45 per cent. Western Digital needs 44.8 per cent of the CPU cycles while transferring data, whereas Seagate needed 28.4 per cent—the lowest in the test.

These drives, if plugged into older machines, will bring them to their knees, and



The Maxtor OneTouch 250 GB

wouldn't allow any CPU-intensive processes such as Photoshop to run while data is being transferred.

To test for real-world scenarios, we copied a single 1 GB image and assorted 1 GB data to and from an external hard drive.

In this test, the Maxtor One Touch was faster than the rest. In the sequential write test, where we copied a 1 GB image from a local drive to the Maxtor One Touch, the transfer took 52.23 seconds. This means that the drive will be very useful to image artists and videographers, because their files usually are a few GB.

The Seagate, despite running at 7,200-rpm with 8 MB of buffer, took 62.78 seconds—which may not make for a major difference if we consider just 1 GB of data